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ORIGINAL RESEARCH

The Impact of Gut Microbiota on Cellular Response to Multidrug Resistant Transporter Substrates *In Vitro* With Focus On P-Glycoprotein

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Abstract :

Pharmacokinetic studies at the host-microbe interface mainly focus on the effects on drug metabolism, and few studies consider the absorption process. We previously showed that transcriptional expression of genes encoding intestinal transporters involved in lipid translocation is altered in germ-free and conventional mice with different bile acid profiles. Therefore, microbial bile acid metabolism, i.e., deconjugation and dehydroxylation of bile acid steroid cores by H. enterobacteriaceae, can affect drug transporter expression and/or activity and alter drug properties. Using a panel of three human intestinal cell lines (Caco-2, T84, and HT-29) that differ in basal transporter expression levels, bile acid binding, and hydroxylation status, several major tributary-encoded genes was shown to affect transcription. Efflux transporter protein. We also investigated whether these effects on transporter mRNA lead to changes in drug properties and activity. The results showed that the binding and hydroxylation state of the bile acid steroid core can influence cellular responses to multidrug resistance (MDR) substrates, but this finding is not related to the direction of gene or protein expression. In particular, we demonstrated that the cytotoxicity of cyclosporine A in the presence of the unconjugated bile acids deoxycholic acid (DCA) and chenodeoxycholic acid (CDCA) in P-gp positive cell lines compared to taurine/glycine conjugated cells. I found that this reflects the following: P-gp in involves a molecular reaction. Overall this study identifies a novel mechanism by which gut microbial metabolites influence drug accumulation and shows that the microbial bile acid unconjugating enzyme, bile salt hydrolase (BSH), possesses this ability. These results suggest that the bile acid species may play a role in improving multidrug resistance through the formation of bile acid species. Accesses and inhibits P-gp-ATPase. The physicochemical properties of nonionization are thought to underpin the preferential ability of unconjugated bile acids to attenuate the efflux of P-gp substrates and sensitize tumorigenic cells to cytotoxic therapeutic agents in vitro. I am. This study provides a new impetus to investigate whether disruption of the gut microbiota and thus the bile acid component of the intestinal metabolome can alter the pharmacokinetics of drugs *in vivo*. These findings may further contribute to the development of less toxic P-gp modulators that can overcome MDR.

Keywords : Bile acid metabolism, Pharmacokinetics, Drug absorption,

Introduction:

The human gastrointestinal tract can be viewed as a microbial ecosystem housing trillions of microbial cells with a diverse functional repertoire. These commensal bacterial assemblages have been widely reported to influence drug metabolism, thereby altering clinical outcomes.

However, relatively few studies have investigated possible effects on the absorption process of drugs. The combination of an "unstirred" water layer, tight junctions, and epithelial cell membranes expressing efflux proteins can, in some cases, severely limit drug transport across the intestinal barrier.

We previously found that the bile acid metabolism of gut microorganisms can influence the solubilization ability of simulated intestinal fluid and, as a result, the solubility of poorly water-soluble drugs (PWSDs) [1].

Bile acids have been extensively studied as emulsifying and permeation enhancers at the millimolar micelle-inducing concentrations used in this previous study and have been shown to have a role in modulating the absorption of lipophilic drugs. However, there is still a shortage. Effects of bile acids on drug transport at micromolar signaling concentrations as intracellular ligands for nuclear receptors, or on other cellular properties that may affect drug transport.

The major human bile acids, cholic acid (CA) and chenodeoxycholic acid (CDCA), are synthesized from cholesterol within hepatocytes. After biosynthesis, these primary bile acids are linked to taurine or glycine, respectively, through a nasylamido linkage in a ratio of approximately 1:3. Because amidation promotes protonation at physiological pH, conjugated bile acids exist in ionized form, more accurately described as salts. Conjugated bile salts are stored and concentrated in the gallbladder over the intermeal period. After a meal, the gallbladder is stimulated by cholecystokinin, and bile salts enter the intestine, where biotransformation by commensal bacteria occurs. First, the microbial enzyme bile salt hydrolase (BSH) catalyzes the hydrolysis of the taurine/glycine tag, releasing unbound free bile acids. The regenerated unconjugated CA and CDCA are suitable for 7α-dehydroxylation mediated by microbial 7α-dehydroxylases and are capable of converting the secondary bile acids deoxycholic acid (DCA) and lithocholic acid (LCA) [2].

All of these unconjugated and dehydroxylated bile acid species are recycled via the portal blood to the liver where taurine/glycine recombination occurs. Although the formation of the bile acid pool is based on synthesis by the host, its composition and physicochemical characteristics are greatly influenced by the microbiome.

In recent years, the role of endobiotic (including bile acid)-inducible nuclear receptors in regulating transporter expression has become clear. Because these nuclear receptors have specific sensitivities to bile acids, subtle changes in local or systemic bile acid profiles can influence host physiology. Previous studies from our group showed that microbial recolonization of germ-free mice significantly affects the transcription of genes encoding lipids and intestinal transporters involved in drug absorption.

Due to the composition of the bile acid pool in habituated mice is chemically more diverse than that of germ-free mice, bile acid metabolism mediated by the gut microbiota may interfere with drug transduction. We hypothesized that this might affect the expression of porter. Drug properties are most commonly influenced by two superfamilies: ATP-binding cassettes (ABCs) and solute carrier transporters (SLCs). Therefore, we focused on elucidating the effects of bile acid uncoupling and dehydroxylation on the transcription and activity of six clinically [2].

Oncogenic multidrug resistance is associated with overexpression of the ABC superfamily of efflux transporter proteins, including P-glycoprotein (P-gp, ABCB1) and breast cancer resistance protein (BCRP, ABCG2). These transport proteins are also abundant in the apical membrane of enterocytes and often act as a barrier to the uptake of sensitive substrates by recycling a portion of the absorbed drug back into the lumen. Downregulation or inhibition of these multiresistance proteins through substrate competition.

Therefore, ATP depletion, or membrane disruption, may increase intestinal absorption and induce tumorigenic chemosensitivity. Some lipophilic drugs, such as verapamil, have been shown to be useful reversal agents *in vitro* and in preclinical animal models but their clinical application is limited by their innate pharmacodynamic likely to be limited by effectiveness. Therefore, it is of interest to investigate endogenous factors that may influence the action of these transporters and identify less toxic reversing agents [3].

Cyclosporine A (CsA) is an immunosuppressive calcineurin inhibitor commonly prescribed in organ transplant settings to prevent or treat allograft rejection. Its clinical use is often complicated by highly variable pharmacokinetics, requiring dose adjustment for each patient. In addition to the susceptibility of CsA to metabolism by CYP3A4, some reports have suggested that differences in bioavailability between patients may be partially due to changes in intestinal P-gp expression. Considering its reported limiting effect on the proliferation of colon cancer cell lines, CsA is considered a clinically relevant P-gp substrate that can induce cytotoxicity in this study.

selected for use in. This latter feature provides an opportunity to assess variations in transporter function as changes in cytotoxicity.

Similarly, SN-38, the active metabolite of the antineoplastic drug irinotecan, was selected as a model substrate to assess the effects of bile acid binding and hydroxylation status on BCRP activity.

Overall, this article highlights the role of bile acid deconjugation and dehydroxylation (controlled *in vivo* by the gut microbiota) in the multidrug resistance phenotype changes of Caco-2 and T84 cell lines, i.e., the intestinal tract. Provides insight into the barrier and in vitro models of the intestinal barrier. One neoplastic colon. First, we investigated the effects of conjugation/unconjugation and trihydroxy/dihydroxy bile acids on messenger RNA (mRNA) transcripts of common enterocyte influx and efflux transporters. Second, we used an in vitro cytotoxicity screen to investigate whether these bile acid changes in transporter mRNAs could cause changes in drug properties and activity. There is no correlation between changes in drug activity and direction of gene protein expression, and given the reported non-transcriptional resistance reversal effect of detergents, it is possible that bile acids influence absorption capacity. Other mechanisms are possible. We investigated the possibility of intestinal epithelium. Because the composition of the gut microbial community, and therefore its metabolites, can vary significantly between individuals, this study provides knowledge of novel mechanisms by which the gut microbiome may contribute to variation in pharmacokinetics. The aim is to expand the base [4].

Materials and methods:

Cyclosporine A (CsA) from Tolypocladium inflatuum (purity ≥ 95%), SN-38 (purity ≥ 98%), verapamil hydrochloride (≥ 99%), Rhodamine 6G, Dimethyl Sulfoxide (DMSO, Hybri-Max, Sterile Filtered, BioReagent, Purity ≥ 99.7%), Sodium Taurocholate Hydrate (≥ 95%), Sodium Cholate Salt Hydrate (≥ 96%), Sodium Taurodeoxycholate Hydrate (≥ 95%), Deoxycholic Acid (≥ 99%), Sodium Taurochenodeoxycholate (≥ 98%), Sodium Chenodeoxycholate (≥ 97%), Thiazolyl Blue Tetrazolium Bromide (MTT), Costar 3412 Polycarbonate Membrane Cell Culture Transwell Insert (24 mm diameter, 0.4 µm pore size), Dulbecco's Modified Eagle Medium (DMEM), MEM Non-Essential Amino Acid Solution (100x), Fetal Bovine Serum (FBS), Penicillin-Streptomycin, Ham Nutrient Mixture F-12 , phosphate-buffered saline, T-175 bottles, tissue culture treatment plates, and other tissue culture reagents were obtained from Sigma-Aldrich, Ireland.

The RNeasy mini kit was purchased from Qiagen Ltd., Ireland. TURBO DNA-free™ kit was purchased from ThermoFisher Scientific, Ireland. Transcriptor reverse transcriptase and Universal Probe Library probes were purchased from Roche LifeScience Ireland [5].

Primer “random”, protector RNase inhibitor, and PCR nucleotide mix were purchased from SigmaAldrich, Ireland. SensiFAST™ Probe No-ROX Kit was purchased from MSC Ltd., Ireland. Primers were purchased from Eurofins Scientific, Ireland. Gels and buffers used for Western blotting were purchased from BioSciences, Ireland. Anti-sodium-potassium ATPase antibody, anti-P-glycoprotein antibody and anti-BCRP antibody were purchased from Abcam, Cambridge, UK. CellTiter Glo and Pgp-Glo assay systems were purchased from Promega, Ireland.

Cell Line and Culture Caco-2 human epithelial colorectal adenocarcinoma cells are maintained by the APC Microbiome Ireland and School of Pharmacy Cell Culture Collection and were originally obtained from the American Type Culture Collection (Rockville) and described herein. was used at passage numbers 36 to 55 in the study [6].

Caco-2 cells were maintained in DMEM supplemented with 10% (v/v) FBS, 1% (v/v) non-essential amino acids, 100 U/ml penicillin and 100 µg/ml streptomycin. T84 human colon cancer cells (lung metastases) obtained from Sigma-Aldrich and maintained in APC Microbiome Ireland cell culture collection were used as passages #5 to 18. T84 cells were maintained in DMEM:F-12 containing 10% (v/v) FBS, 100 U/ml penicillin and 100 µg/ml streptomycin. In the studies described here, human HT-29 colon adenocarcinoma cells maintained by APC Microbiome Ireland were used at passages #13 to 25. HT-29 cells were maintained in McCoy's 5A supplemented with 10% (v/v) FBS, 100 U/ml penicillin, and 100 µg/ml streptomycin. Cell lines were routinely grown in 175 cm2 tissue culture flasks, passaged twice weekly at 1:5 dilution and incubated at 37 °C in a humidified chamber containing 5% CO2 and 95% air.

RNA extraction, cDNA synthesis, and quantitative PCR:

Aco-2, T84, or HT-29 cells were seeded in 6-well plates at a density of 50,000 cells/cm2 with supplemented growth medium (2.5 ml/well). I multiplied it. After 24 h, cells were incubated with 100 µM (Caco-2 and HT-29) or 50 µM (T84) of the test bile acids, or DMEM:F-12 (Caco-2 and) containing an equal volume of DMSO (vehicle control).

I processed it with T84) or serum-free DMEM (HT-29). After 24 h of incubation, the treatment medium was aspirated, cells were washed once with phosphate-buffered saline (PBS), and RNA was isolated using the RNeasy Mini kit (Qiagen) according to the manufacturer's instructions. The quality and quantity of purified RNA (and TURBO-DNase treated) RNA was analyzed using a NanoDrop spectrophotometer. Samples were stored at -80°C until the next step described below [7].

cDNA synthesis was performed using 300-500 ng of total RNA isolated from cells with 5x transcription buffer (Roche), random primers (Roche), transcription reverse transcriptase (RT) enzyme (Roche), and protector RNase inhibitor (Roche) and dNTPs (Roche) in a total volume of 20 µL. cDNA was synthesized using a thermal cycler according to the following protocol steps: 1) primer annealing at 25°C for 10 min, 2) cDNA synthesis at 55°C for 30 min and 3) 5 min at 85°C. Heat inactivate at RT for 4 min and then cool to 4°C. The reaction mixture was then diluted to 100 µL with nuclease-free H2O and stored at -20 °C until analysis. Specific primers for each gene designed using the Universal Probe Library (Roche) were used for quantitative PCR (qPCR). One cycle consisted of 95°C for 10 s, 60°C for 45 s, 70°C for 1 s, and repetitions, for a total of 55 cycles. The expression of target genes in each sample was assessed using cross-threshold (Ct) values. Results were normalized to the β-actin gene, and relative mRNA expression was calculated based on the comparative cycle threshold method.

Cell viability assay:

Cell viability was assessed using the CellTiter-Glo luminescent cell viability assay (Promega), which can determine the number of viable cells in culture based on ATP quantification. Cells were exposed to conjugated unconjugated bile acids or primary secondary bile acids in the absence and presence of model efflux substrates (such as SN-38 and CsA) to determine cell viability and drug response. We investigated the effects of microbially mediated bile acid metabolism [8].

cells were plated in 96-well plates at a density of 1.6 × 104 cells per well and cultured with 100 µL of additional growth medium for 24 h. After aspirating the growth medium, cells were exposed to serum-free DMEM:F-12 (Caco-2 and T84) or serum-free McCoy's medium (HT-29) containing test bile acids and/or SN-38 or CsA. At the concentrations.

All test substances were dissolved in DMSO at a concentration of 100 mM or 10 mM for bile acids and SN-38/CsA and then diluted in DMEM:F-12 (Caco-2 and T84) or serum-free McCoy's medium . Depending on the concentration or combination of test substances, medium containing 0.05-0.2% DMSO was used as a negative control. After a 24-hour treatment period, the luminescence of the plates was measured using a BioTek Synergy 2 multimode microplate reader [9].

A tetrazolium-based colorimetric MTT assay was used to verify the noncytotoxicity of 100 µM unconjugated bile acids. Similar to the CellTiter Glo assay, Caco-2 cells were seeded at a density of 1.6 × 104 cells per well in 96-well plates in supplemented DMEM growth medium. After 24 hours, the medium was removed and cells were treated with 100 µM test bile acids in DMEM:F-12 or an equal volume of DMSO vehicle (control). 20 µL of a 5 mg/mL solution of MTT (3-(4,5-dimethylthiazole-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide) [3].

Three hours before the end of the 24-hour treatment period. Now phosphate buffered saline (PBS) was added to each well and the cells were incubated for an additional 3 hours. The medium was then removed and cells were washed twice with PBS. Solubilization of formazan crystals was achieved by adding 200 µL of DMSO to each well. The optical density (OD) was measured with background subtraction at 570 nm using a VICTOR2™ 1420 multilabel counter. Viability was calculated relative to vehicle-treated cells and corresponds to 100% viability. All viability data are presented as the mean ± SEM of three to four independent experiments performed in triplicate [10].

Western blot analysis:

Cells were treated with 100 µM (Caco-2 and HT-29) or 50 µM (T84) TDCA, DCA, or DMSO vehicle control for 24 h as described above for RNA isolation. After treatment, the growth medium was aspirated and cells were rinsed with cold PBS. Cells were then cultured in RIPA buffer (50 × mM Tris (pH 8.0), 150 mM NaCl, 1% Triton). Culture plates containing lysis buffer were left on ice for 10 min, after which the lysates were transferred to a pre-chilled Eppendorf. Cells were removed using a cell scraper tubes were collected, and lysates were kept on ice for an additional 20 min. To remove debris, the lysate was centrifuged at 14,000 rpm for 15 min at 4°C and the supernatant was collected. Total protein content was determined according to the BCA assay (Pierce). Immunoblotting were denatured by boiling for 10 min.

Samples used for Na+/K+-ATPase (loading control) and P-gp immunoblotting were not denatured. Total protein (13.5 µg) was separated on a 3-8% Tris-acetate gel (NuPAGE) using NuPAGE Tris-acetate SDS running buffer. 500 µL of Bolt antioxidant was added to the cathode chamber. After electrophoresis (150 V until sufficient resolution was achieved), proteins were transferred to a PVDF membrane (0.45 µm) (30 V for 90 min). After transfer, the membranes were blocked with 5% (w/v) nonfat dry milk in PBS-0.1% (v/v) Tween-20 for 1 hour at room temperature. The membranes were then incubated with the relevant primary antibodies in 15 ml of PBS containing 1% w/v bovine serum albumin and 0.1% v/v Tween-20 overnight at 4°C with gentle shaking. Primary antibodies were used at dilutions of 1:10,000, 1:1,000, and 1:1,000 for Na +/K + -ATPase, BCRP, and P-gp, respectively. The next day, membranes were incubated with goat anti-rabbit HRPase-conjugated secondary antibodies diluted 1:10,000 in 5% (w/v) nonfat dry milk in PBS-0.1% (v/v) for 1 h at room temperature.) Tweens were cultured. 20. Bands were visualized using the Advansta WesternBright Quantum kit. Protein expression levels were quantified using NIH Image J software and the relative amounts of BCRP and P-gp protein expression relative to Na+/K+-ATPase were calculated [11].

Rhodamine 6G Accumulation Assay:

P-gp activity of conjugated (TCDCA and TDCA) and unconjugated bile acids (CDCA and DCA) was evaluated using rhodamine 6G (Rho 6G) in Caco-2 and T84 cells. was determined by measuring the intracellular accumulation of. Test with or without the simultaneous presence of bile acids. Briefly, cells were incubated with 5 µM(Caco-2) or 2.5 µM (T84) Rho 6G (in DMEM:F-12) for 3 h in the presence or absence of 100 µM test bile acids.

After the incubation period, cells were washed twice with serum-free medium and then lysed with 100 µL of 1% (wt/vol) Triton X-100 (Sigma-Aldrich, Ireland) at room temperature. Intracellular levels of Rho 6G were spectroscopically analyzed using a VICTOR2™ 1420 multilabel counter with excitation (λEX) and emission (λEM) wavelengths set at 485 nm and 615 nm, respectively. Quantified fluorometrically. Data were expressed as percentage accumulation of Rho 6G in test cells at a ratio of To accumulation in untreated control cells, arbitrarily set to 100%. Increased intracellular accumulation of Rho 6G corresponds to decreased efflux activity.

To gain insight into the mechanism by which bile acids regulate the increased cellular uptake of Rho 6G, accumulation assays were performed at 4°C, 21°C and 37°C as described above. Experimental data are expressed as the mean ± SEM of four independent experiments (n = 6 replicates per experiment) [3].

Rhodamine 6G Apparent Permeability Test Bidirectional permeability evaluation was performed using polycarbonate membrane cell culture transwell inserts (24 mm diameter, 0.4 µm pore size). Caco-2 cells were seeded in supplemented DMEM at a density of 63,000 cells/cm2 and differentiated over 21-23 days until a transepithelial electrical resistance (TEER) of 350 ohms or higher was reached.

TEER values were measured using an EVOM epithelial volt-ohmmeter (World Precision Instruments, USA) at the beginning of cell culture medium exchange (every other day) and periodically before and after testing. During maintenance and testing, the volumes of the apical and basolateral chambers were 1.5 ml and 2.6 ml, respectively [12].

All transport experiments were performed in triplicate. On test day, the average TEER value after switching to the test medium (DMEM:F-12) was 408 ± 12 ohms. In the test conditions, 100 µM test bile acids (TDCA/DCA) or verapamil (positive control) were added to both sides of the monolayer 1 h before the start of the transport assay.

After this 1-h preincubation period, 5 µM Rho 6G (from a 5 mM DMSO stock solution) was added to the donor chamber, and drug transport to the recipient chamber was analyzed for 3 h at 37°C. After 30, 60, 90, 120, 150, and 180 min, 100 µL samples were removed from the recording chamber and replaced with fresh medium (containing test bile acids or verapamil as appropriate). First a 100 µL sample was also collected from the donor chamber (supplemented with an equal volume of 5 µM Rho 6G-containing medium) to confirm the initial donor concentration.

P-gp-ATPase activity test:

Confirmation of the effect of test bile acids on P-gp-ATPase and P-gp substrate activity on P-gp-ATPase is performed using the P-gp-Glo test system “Promega”. was tested. The manufacturer's protocol was followed with some modifications (three hour incubation time was used due to the low basal ATPase activity of the P-gp membrane). All data are presented as mean ± SD (n = 3).

Statistical analysis:

A two-tailed independent samples t-test assuming normal distribution and equal variances was used to assess the statistical significance of two-sample comparisons [1]. A P value < 0.05 was considered statistically significant. All statistical analyzes were performed using GraphPad Prism 5 (San Diego, CA).

Results:

Effect of bile acid binding and hydroxylation status on the expression of drug transporters in the intestine:

In the liver, bile acid-dependent activation of FXR induces bile acid excretion It is known to induce porter expression. Bile salts, cholesterol, and phospholipids in the bile ducts. Although there is evidence that bile acids can affect the expression of various ABC transporters in non-hepatic tissues to our knowledge, bile biotransformed by microorganisms The effects of acid on enterocyte drug transporters and drug activity have not yet been considered. Therefore, we investigated the effects of microbial BSH and 7α-dehydroxylase activities on the expression of six intestinal transporters that are often involved in altering pharmacokinetics *in vivo*, either as binding/unbinding or as primary secondary was investigated in three intestinal cell lines. Simulated bile acids. The bile acid concentrations used were found to be noncytotoxic to the cell lines and were within the range expected in fecal water of healthy individuals after a normal diet (figure 1) [13].

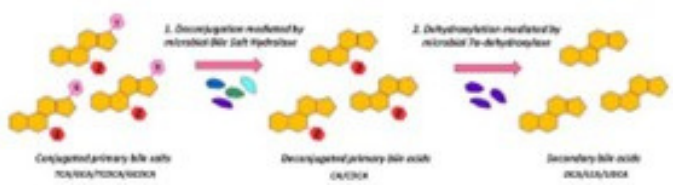

Figure 1: Schematic overview of microbial bile acid metabolism.

In general, unconjugated DCA and CDCA increased the expression of genes encoding enterocyte drug transporters compared to vehicle controls and their conjugated counterparts. However, the estimated effects of microbial bile acid unconjugation on ABCB1 (P-gp) and ABCG2 (BCRP) expression differed from this generalizable trend across the three cell lines.

Comparison of the expression profiles of CA- and DCA-treated cells was consistent with the bile acid unconjugation results. In short, increased hydrophobicity of the steroid core of DCA was associated with increased expression of intestinal drug transporter mRNA. 3 cell lines. Deviations from this general trend were also observed for ABCB1 (P-gp) and ABCG2 mRNA (BCRP). To examine whether unconjugated dihydroxy bile acids are likely to upregulate drug transporter mRNA through stimulation of FXR, we evaluated their effects on the expression of FGF19, a known FXR-regulated gene. In general, unbound DCA and CDCA increased FGF19 mRNA in a manner similar to ABCC2 and ABCC3, which are reported to be regulated by FXR.

Effects of bile acid unconjugation on cellular responses to therapeutic agents:

To examine whether the observed changes in intestinal drug transporter expression translate to changes in pharmacokinetics and activity in vitro, Caco- The first study was conducted on 2. - Performed cell lines. DCA was found to increase the cytotoxicity of CsA and SN-38, known substrates of P-gp and BCRP-mediated efflux, respectively. This ability of DCA to enhance the cytotoxicity of these MDR substrates did not correlate with ABCB1 (P-gp) or ABCG2 (BCRP). Therefore, considering that the roles of P-gp and BCRP in limiting the intestinal absorption of many clinically important therapeutics are well-studied, we investigated the underlying mechanisms. I tried to find out more. For this purpose, we used three cell lines with different P-gp and BCRP phenotypes. Uncoupling of the bile acid steroid core differentially modulates SN-38 and CsA cytotoxicity in a manner opposite to that expected based on ABCG2 (BCRP) and ABCB1 (P-gp) expression This has been shown. DCA, but not Tauro conjugate, significantly decreased Caco-2 cell viability (69.4 ± 2.07%) compared to SN-38 alone (105.10 ± 1.86%, P = 0.0002. Bile acid unconjugation did not affect the sensitivity of T84 or HT-29 cells to SN-38 (figure 2).

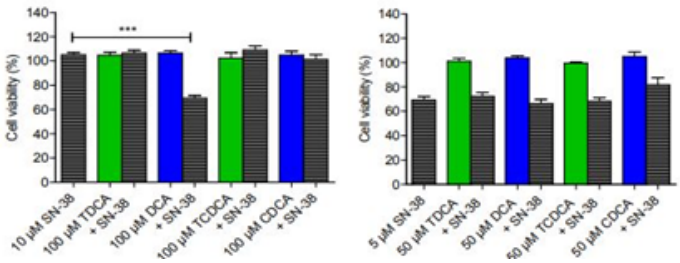

Figure 2: The effect of bile acid deconjugation.

DCA was found to enhance the cytotoxicity of CsA against Caco-2 cell line, with this combination yielding 59.52 ± 2.71% cell viability, whereas CsA alone yielded 94.88 ± 1.66%. (P = 0.0004). CDCA was also found to increase the cytotoxicity of CsA. After dual treatment, a decrease in cell viability of 64.86 ± 2.70% vs. 94.88 ± 1.66% was observed compared to CsA treatment alone (P = 0.0007). These observations could also not be explained by bile acid-induced changes in Caco-2 ABCB1 (P-gp) mRNA transcripts. The glycine-conjugated bile salts behaved similarly to the taurine conjugates. These findings were further confirmed in the T84 cell line, which is also P-gp positive. The HT29 cell line, which is known to be P-gp negative, served as a control and showed no significant changes in drug response (figure 3).

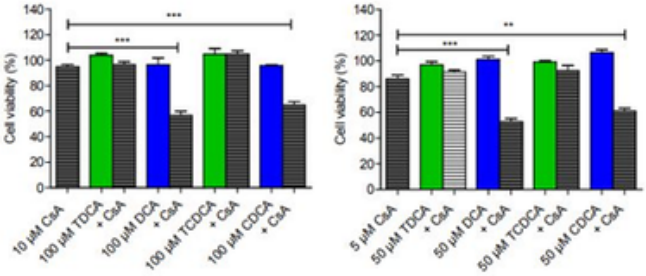

Figure 3: Bile acid- on Caco-2 (a), T84 (b) and HT-29.

Effect of bile acid dehydroxylation on cellular responses to therapeutic agents:

Simulating the effect of microbial 7α-dehydroxylation of the cholate steroid ring on the propensity of bile acids to adapt to the cytotoxic effects of SN-38 and CsA. I did it. Similar to the effect of deconjugation, dehydroxylation of bile acids was associated with increased drug activity, as shown by decreased cell viability after replacing CA with DCA Co-treatment of Caco-2 cells with SN-38 and DCA resulted in cell viability of 61.56 ± 3.47%, compared to 90.52 ± 7.8% for SN-38 alone (P = 0,0147) [14]. Bile acid hydroxylation status did not affect SN-38 cytotoxicity in the BCRP-negative T84 cell line and HT-29 cell line (Figure 4).

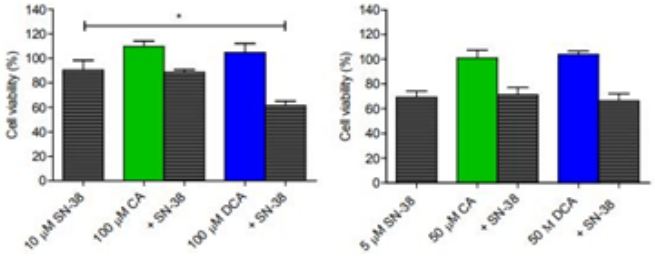

Figure 4: Bile acid dehydroxylation on SN-38 in the Caco-2 cell line.

DCA but not CA was also shown to enhance the cytotoxicity of CsA (43.85 ± 2.47% after combination treatment compared with 91.24 ± 14% for CsA alone (P = 0.0157). This effect was confirmed in the P-gp-positive T84 cell line. Cell viability was not affected in the HT-29 cell line. Similar to the effects of uncoupling, these results generally cannot be correlated with transcriptional effects on ABCB1 (P-gp) and ABCG2 (BCRP). Overall, the results suggest that unconjugated bile acids and secondary bile acids are better able to modulate the uptake activity of common efflux substrates (figure 5) [2,13].

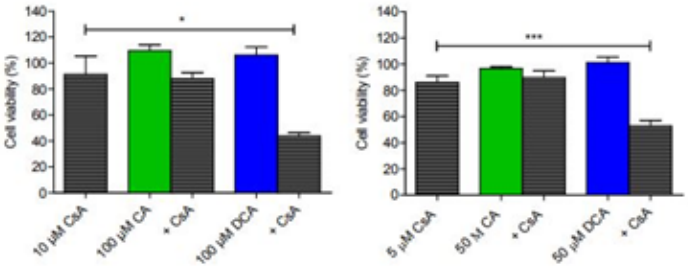

Figure 5: Bile acid dehydroxylation on HT-29 cell line.

Effect of bile acid conjugation status on rhodamine 6G uptake and bidirectional transport:

Unconjugated bile acids are more capable of inhibiting intracellular accumulation of P-gp substrates by inhibiting efflux activity. To further investigate the hypothesis, transport studies were performed using a range of uptake and bidirectional transport fluorescent rhodamine dyes, Rhodamine 6G (Rho 6G). Rho 6G was identified as a P-gp substrate based on its ability to stimulate basal P-gp ATPase activity. In the Caco-2 cell line, after 3 h co-incubation at 37 °C, unconjugated DCA and CDCA increased the intracellular accumulation of HisRho 6G, but not the Tauro-conjugated counterpart. CsA 100 µM, used as a positive control to validate the assay, significantly increased the intracellular accumulation of Rho 6G (data not shown). Although not statistically significant, a similar trend was also observed in his T84 cell line. The less pronounced phenotype observed in T84 cells may reflect the reduced level of P-gp expression in this cell line.To further demonstrate that the increased Rho 6G uptake observed in the presence of unconjugated bile acids occurred through a transporter-mediated energy-dependent process, parallel experiments were carried out at 4°C and at room temperature (ca°C).

Accumulation of Rho 6G in the absence and presence of test bile acids did not differ at 4°C or 21°C. This suggests that an energy-dependent process is responsible for the enhancement by unconjugated bile acids at 37°C (figure 6) [4].

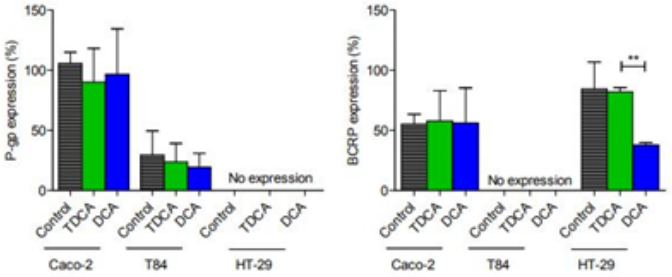

Figure 6: Representative western blots of P-glycoprotein**.**

Exploring the mechanism by which bile acids inhibit P-glycoprotein function:

P-gp functions as an ATP-dependent drug efflux pump. Compounds that interact with P-gp can be classified as either substrates or inhibitors, depending on their propensity to stimulate or inhibit the transporter's ATPase activity (i.e., its ability to hydrolyze ATP). To elucidate the possible mechanisms underlying the enhanced CsA toxicity and Rho 6G accumulation observed in the presence of unconjugated bile acids, the effects of the test bile acids on P-gp ATPase activity were evaluated using artificial investigated using human P-gp membrane vesicles. In general, both conjugated and unconjugated bile acids inhibited basal P-gp-ATPase. Test bile acids were also shown to inhibit P-gp-ATPase stimulated by substrates (verapamil or Rho 6G), although this was not a generalizable effect (figure 7) [9].

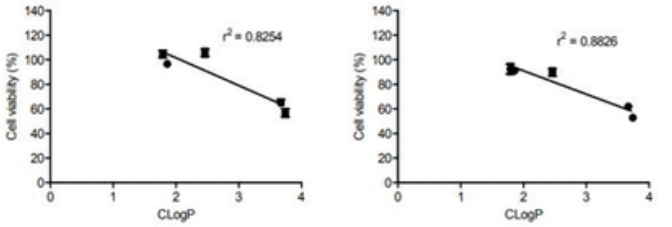

Figure 7: impact of bile acids on CsA-induced cytotoxicity toward the Caco-2.

It is unclear what percentage of the treated bile acid concentration (100 µM) is available for interaction with P-gp in the cell line, we also investigated the effects of test bile acids at a concentration of 5 µM. The tested bile acids at a concentration of 5 µM were also found to inhibit his P-gp-ATPase activity [7, 14].

Discussion:

Effect of bile acid binding and hydroxylation status on the expression of genes encoding drug transporters:

Several recent studies have shown that the intestinal Attempts have been made to investigate the role of the internal microbiome. Previous studies by our research group using germ-free and conventional mice with different bile acid profiles identified Transcriptional changes in intestinal transporters involved in lipid transport. Reanalysis of this dataset further revealed altered transcription of enterocyte transporters involved in drug translocation, Supporting Information). To our knowledge, the influence of microbial-mediated bile acid conversion on the expression of human intestinal drug transporter is not yet clear.

A panel of three human intestinal cell lines was used to study the effects of individual bile acids on the transcriptional expression of drug transporters. Because the properties of both endogenous molecules, such as bile acids, and drugs are most commonly associated with two superfamilies: ABC transporters and SLC transporters, we have identified several clinically relevant members. We have focused on elucidating the effects of bile acid conversion on the transcription and activity of. In general, unconjugation and dehydroxylation of bile acids was associated with transcriptional upregulation of genes encoding these transporters (consistent with mouse internal data).

The-unconjugated dihydroxy bile acid signature is the basis for the induction of common human intestinal transporters, as exemplified by the ability of dihydroxy-CDCA and DCA, but not trihydroxy-CA, to commonly upregulate mRNA transcripts. These results may at least partially explain recent reports of altered expression of drug transporters in dysbiotic (sterile and antibiotic-induced) animal models.

The expression of drug transporters is an important determinant of local and systemic drug exposure and resulting pharmacodynamic effects. Therefore, interindividual differences in drug response may be related to changes in transporter expression. Considering the results in gnotobiotic mice and cultured intestinal cells, it is suggested that disruption of the gut microbiota, and thus of the bile acid pool, may affect pharmacokinetics and warrants further investigation.

Such changes in transporter gene expression may be determined by bile acid-specific stimulation of nuclear receptors such as FXR, PXR, and constitutive androstane receptor (CAR). In support of this statement, we demonstrated that unbound DCA and CDCA were regulated by FGF19. However, the mechanisms underlying the differential expression of ABCB1 (P-gp) and ABCG2 (BCRP) in the three cell lines after bile acid treatment have not yet been conclusively elucidated. We also hope to decipher which of these commonly used cell lines provides the most relevant information to the *in vivo* environment and which are the best candidates for transcriptional studies.

They reported similar correlation coefficients between all three cell lines and human small intestine (ileal) tissue biopsies: 0.672 (Caco-2), 0.708 (T84), and 0.679 (HT-29). However, the best correlation was observed with biopsies of human colon tissue and T84 and -HT-29 cell lines. It is also important to note that mRNA gene expression data does not necessarily correlate with the expression pattern of the encoded protein.

Effects of bile acid binding and hydroxylation status on the cytotoxicity of multiresistant transporter substrates:

Bile acids can have a profound effect on drug transporter mRNA expression in mouse and cultured human intestinal cells Based on this knowledge, we considered possible effects on bile acids.

Drug absorption using in vitro cytotoxicity screening of drugs. Changes in drug toxicity, which should follow changes in cellular accumulation, were used as preliminary markers to evaluate the effects of treatment with bile acid on the absorption and activity of a series of drugs.

Among the bile acids tested, unconjugated dihydroxy bile acids modulated the cytotoxicity of model MDR substrates SN-38 and CsA, but these effects appeared to be drug-specific. We hypothesize that the transporter specificity of these substrates (BCRP and P-gp, respectively) may explain the distinct propensity of modulate the cytotoxicity of unconjugated dihydroxy bile acids. However, such bile acid-dependent drug toxicity may be due to bile acid-induced changes in ABCG2 (BCRP) and ABCB1 (P-gp) mRNA, as well as changes in BCRP or P-gp. It cannot be explained by expression changes. -GP. gp protein.

Since drug efflux is an energy-dependent process, cellular energy consumption is a source of inhibition for multidrug resistance. Neither conjugated nor unconjugated bile acids caused significant changes in cellular ATP levels at the concentrations used.

Therefore, depletion of intracellular ATP by bile acids may be the basis for chemosensitization of Caco-2 and T84 cells, as previously reported for non-physiological surfactants such as Pluronic P85. There is a need to sensitize cells to SN-38 and CsA through alternative functional effects on these transport proteins.

Effect of bile acid binding status on cellular uptake of P-gp substrates:

P-gp activity in cell lines is routinely tested by monitoring the accumulation and efflux of small fluorescent molecules such as rhodamine dyes. Rho 6G has been reported to be a relatively hydrophobic P-gp substrate and was confirmed here by its ability to stimulate basal P-gp ATPase.

We show that the P-gp transporter plays a role in limiting the intracellular accumulation of Rho 6G in Caco-2 cells, with little or no contribution from multiple resistance protein isoforms.

Unconjugated DCA and CDCA significantly increased Rho 6G accumulation in Caco-2 cells ($P < 0.05$). This effect was not evident in the presence of Tauro-binding equivalents. Parallel studies performed at 4°C and 21°C confirmed that the effect of bile acid unconjugation on Rho 6G accumulation is an energy-dependent process.

A similar trend, although not statistically significant, was observed in T84 cells. This phenomenon supports the hypothesis that unconjugated bile acids can positively regulate the accumulation of P-gp substrates and raises the possibility of a role for microbial BSH in influencing the uptake and activity of molecules susceptible to P-gp exclusion.

It is pointed out that it suggests sex. We note reports in non-intestinal cell lines suggesting that Rho 6G, in addition to its sensitivity to P-gp efflux, may be susceptible to uptake by organic cation transporters (OCT1 and OCT2).

Although basal mRNA expression of SLC22A1 (encoding OCT1) is low in the intestinal cell lines used in this study, the contribution of these transporters to Rho 6G warrants further investigation.

Effect of bile acid conjugation status on the transport of P-gp substrates across Caco-2 monolayers: a model of the gastrointestinal barrier:

Effect of bile in bidirectional transport studies (using polar Caco-2 monolayers) Performed acid unconjugation in Rho 6G translocation. This allows for the evaluation of changes in apparent permeability, especially for absorption and secretion. Bile acids were added to both the apical and basolateral compartments since they are present in both the luminal and portal circulation *in vivo*. This experiment is the first in the literature to demonstrate that micromolar concentrations of DCA (compared to its tauroconjugate) can preferentially reduce the basolateral to apical secretion of Rho 6G, suggesting a possible role for microbial BSH in regulating P uptake -gp substrate. It was previously demonstrated that 1.2 mM DCA reduced the secretion of epirubicin, a P-gp substrate, across Caco-2 cell monolayers. speculated that he was induced by DCA. It was confirmed that membrane disruption may change membrane fluidity and inhibit P-gp activity. Given the surfactant nature of bile acids and the observed decrease in TEER, changes in membrane fluidity may partially influence the ability of certain bile acids to avoid Pgp-mediated efflux. 4.5 Effect of bile acid binding status on P-gp-ATPase activity.

ABC transporters contain an ATP-binding domain with ATPase activity. Ability to hydrolyze ATP to ADP (adenosine diphosphate), thereby providing energy for substrate transfer. Many P-gp modulators, such as verapamil and CsA, are themselves substrates and competitively inhibit its function without interfering with the catalytic turnover of the pump. Unbound DCA and CDCA were found to modulate CsA cytotoxicity, but neither bile acid caused changes in basal P-gp). Therefore, it was hypothesized that unconjugated bile acids could be substrates of P-gp efflux or inhibitors of P-gp-ATPase, thereby preferentially attenuating active efflux both conjugated and unconjugated bile acids were found to generally inhibit P-gp-ATPase activity stimulated by substrates. The inability of the P-gp-Glo assay system to identify the role of bile acid binding status in the differential regulation of P-gp-ATPase activity most likely reflects the artifactual nature of membrane vesicles. Physiological membranes are relatively impermeable to ionized (bound) bile salt species and therefore their transport is dependent on carrier proteins.

In contrast, a significant portion of unconjugated bile acids exists in unionized form at intestinal pH values and is therefore available for passive nonionic diffusion.

In general, the unionized form of weak acids penetrates the intestinal membrane much faster than the ionized form. In cellular models, the cytosolic leaflet of the plasma membrane, which contains the substrate binding site and two nucleotide binding domains, must be reached by passive diffusion of molecules through the lipid bilayer, whereas the plasma membrane fraction (We propose that unconjugated bile acids such as DCA and CDCA are most likely to interact with P-gp in cellular systems due to their increased hydrophobicity. This is the first study to report the ability of bile acids to modulate P-gp activity by inhibiting P-gp-ATPase. However, it has been shown that glycocholic acid (GCA), taurocholic acid (TCA), and CA can significantly reduce basal BCRP (ABCG2) ATPase activity. The bile acids included in our study were reported by Telbisz et al. Although not considered, differential inhibition of BCRP-ATPase by DCA such as compared to TDCA may partially explain and require the increased cytotoxicity of the BCRP(ABCG2) substrate SN-38. This suggests that there is a possibility. Overall, the possibility that bile acids compete with P-gp substrates for competitive extrusion across the active site was ruled out, as bile acids tend to inhibit rather than stimulate the basal P-gp ATPase.

Conclusion:

Consistent with internal observations of changes in intestinal transporter mRNA transcripts in germ-free and conventionalized mice, we found that unconjugated/dihydroxy bile acids were isolated from genes encoding human drug transporters in vitro. It was shown that the expression of Future studies should focus on investigating possible effects on pharmacokinetics *in vivo*. This study revealed that the binding and hydroxylation state of bile acids modulate the uptake of efflux transporter substrates and thereby determine their ability to avoid multidrug resistance in vitro. Therefore, this study indirectly identified the role of microbial bile acid metabolism in regulating MDR transporters. In the case of P-gp, this appears to be mediated by inhibition of ATPase activity. The generalizable attenuating effect of unconjugation on P-gp function may be attributed to the more favorable physicochemical properties of non-ionization, suggesting a structure-activity relationship between the bile acid steroid core and reversal of multidrug resistance.

Future studies should clarify whether this phenomenon can persist in more complex *in vivo* environments in the presence of fat-soluble vitamins and lipophilic xenobiotics. This study provides further impetus to consider microbial metabolites in addition to the host as factors that can influence drug pharmacokinetics. Disruption of the human gut microbiota and thus the bile acid metabolome may contribute to altered drug absorption across the intestinal barrier and thus potentially be a source of pharmacokinetic variations.

Conflict of interest:

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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